

A Report on Equitable Expansion of Federally Qualified Health Centers and Optimization-Based Decision Support

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Abstract

This report presents a spatial optimization framework for equitable expansion of Federally Qualified Health Centers in Chicago. We document four critical gaps in the current system: the absence of location evaluation tools for decision makers, an obsolete underservice scoring formula (IMU), a flawed MUA designation process, and unmapped referral networks despite federal mandates. We describe how optimization and data science can address each one. We report on the data infrastructure and the interactive dashboard we have been developing, and outline what the optimization model can deliver: district and superdistrict design, candidate location analysis, and redesigned underservice metrics. The report includes a roadmap for the next steps.

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1 The Problem and Motivation

Chicago’s primary care safety net is operated largely through Federally Qualified Health Centers (FQHCs): community-based clinics that receive federal funding through HRSA’s Health Center Program to serve low-income and uninsured populations regardless of ability to pay. In Chicago, 25 parent organizations operate 166 primary care FQHC sites (cleaned from 213 total sites by removing shelters, school-based outreach, and administrative offices), managed under 1,363 grants awarded by HRSA’s Bureau of Primary Health Care between 2016 and 2026 [9]. Nationally, in 2024 more than 32.4 million people used HRSA-funded health centers through 139.4 million total visits; about 90% of these patients had incomes at or below 200% of the federal poverty level, including 25.1 million uninsured, Medicaid, and Medicare patients [9]. Federal appropriations have grown from \$554 million in 1990 to over \$6 billion today [6]. Since 2014, the number of medically underserved people has nearly doubled from 56 million to over 100 million; a study commissioned by NACHC found that over 100 million people in the United States may not have access to a usual source of primary care due to a shortage of providers in their community [15].

HRSA regulates FQHCs through the Health Center Program Compliance Manual [3], which imposes requirements that are inherently spatial planning problems:

- R1. Catchment area design** (Ch. 3): Health centers must define and annually review the boundaries of the catchment area to be served such that the area’s size ensures services are available and accessible to the residents of the area promptly; boundaries must eliminate, to the extent possible, barriers resulting from the area’s physical characteristics, its residential patterns, its economic and social groupings, and available transportation.
- R2. Needs assessment** (Ch. 3): Centers must assess unmet need using available health resources in relation to the size of the area and its population, including appropriate ratios of primary care physicians, along with health indices (infant mortality), economic factors (poverty rate), and demographic factors (population 65+), the exact four components of the IMU score.
- R3. Accessible locations** (Ch. 6): Sites must be accessible to the patient population relative to where this population lives or works, considering access barriers and distance and time taken for patients to travel to or between service sites.
- R4. Hospital referral relationships** (Ch. 8): Each center must develop an ongoing referral relationship with one or more hospitals to assure continuity of care within the catchment area.
- R5. Collaborative relationships** (Ch. 14): Centers must establish and maintain collaborative relationships with other health care providers within the catchment area, local hospitals, and specialty providers to reduce the non-urgent use of hospital emergency departments.

Each of these requirements involves spatial decisions such as where to draw boundaries, how to measure accessibility, which hospitals to partner with, yet HRSA provides no quantitative tools for making them. Health centers are left to determine these on their own, and the results are uneven. Below we identify four critical gaps in the current system.

Gap 1: Decision makers have no way to evaluate a location. When an organization applies for an HRSA New Access Point grant, it must justify *why this location*. But there is no tool to answer that question rigorously. How many underserved residents are within 30 minutes by transit? What is the current capacity of nearby facilities? Will the new site relieve overloaded neighbors or duplicate existing coverage? Nationally, Chang et al. [4] showed that new FQHCs certified between 2007 and 2014 were much less likely to be in high-poverty areas than existing ones (20.5% vs 31.4%).

The current system responds to where grant applications originate, not where need is greatest.

Gap 2: The IMU score is outdated. HRSA designates Medically Underserved Areas (MUAs) using the Index of Medical Underservice (IMU), a composite of four indicators: provider-to-population ratio, infant mortality, poverty rate, and percentage aged 65+. Each is converted to a 0 to 25 scale and summed. The total IMU score ranges from 0 to 100, where lower values indicate greater underservice. Areas scoring $IMU \leq 62$ qualify as medically underserved. But this formula dates from the 1970s. Ricketts et al. [18] argued nearly 20 years ago that the method no longer adequately captures modern healthcare access barriers. It omits insurance coverage, a primary barrier to care. It omits transportation access, often the main factor in whether a clinic is reachable. It omits chronic disease burden (diabetes, hypertension, obesity), the conditions FQHCs actually treat. And it uses a hard binary cutoff: two neighboring tracts with IMU scores 63 and 62 are considered not eligible and eligible, respectively. Moreover, a tract scoring 61 and a tract scoring 30 are treated identically, even though their needs are vastly different.

Gap 3: The MUA designation process has significant limitations. To obtain an MUA designation, an organization must submit an application to HRSA proposing a rational service area, typically a group of census tracts. The applicant draws the boundary, not HRSA [3]. HRSA then computes the IMU score for the proposed area. This means boundaries can be *gerrymandered* to push IMU below 62, there is no systematic process to determine which groupings of tracts *should* be designated, and communities without an organization capable of navigating the federal application never get designated at all. Ricketts et al. [18] argue that this applicant-driven process systematically excludes communities that lack an organization capable of navigating the federal designation system. Milano [14] found a similar pattern statewide in New Jersey: 1,139 census tracts scored medium or high risk on a composite access index, but only 248 held MUA designation. In Chicago, Kim et al. [11] found that 87 census tracts meet IMU eligibility but lack formal designation, predominantly on the South Side. Between 2009 and 2012, 12 clinics closed while 68 opened, yet MUA-eligible but undesignated tracts received only 3 new FQHCs compared with 41 in designated tracts. The resulting FQHC rate in undesignated tracts was only 15.5 per 100,000 poor, versus 40.8 in designated areas. The designation gap is built into the process.

Gap 4: HRSA requires referral agreements but provides no network analysis. Ch. 8 mandates that each health center develop an ongoing referral relationship with one or more hospitals. Ch. 14 requires collaborative relationships with local hospitals and specialty providers to reduce the non-urgent use of hospital emergency departments. But which hospital should a given clinic partner with? Are there communities where patients have no viable referral path to specialist care? Research shows only 34.8% of referrals in the U.S. result in completed specialist visits [16], and the failure rate is worse in underserved areas. No one has mapped the actual referral network for Chicago to identify where these breakdowns occur.

These gaps are connected. In 2005, the Chicago Department of Public Health [19] documented that the ratio of safety net capacity to medically needy population varied *tenfold* across regions, from 0.4 in the Northwest to 5.1 in the Central region. Of 484,120 low-income uninsured Chicagoans, only 35% received community-based care; 312,395 had no medical home. The South region saw visits *decline* even as the number of sites increased, suggesting that *where* facilities are placed matters as much as *how many*. Two decades later, HRSA still tracks grants at the organization level, not the site level, with no public mechanism to evaluate whether funding reaches the neighborhoods with the greatest need. Pérez et al. [17] found through optimization modeling that more funding does not always lead to better health outcomes: more sites do not help if they are in the wrong places.

People in poor, predominantly Black neighborhoods on Chicago’s South Side have a life expectancy 30 years shorter than those in affluent neighborhoods nine miles away, the largest such gap in the United States [20]. Luo et al. [13] showed on the same South Side that better spatial accessibility to primary care is associated with lower hypertension prevalence. Lee et al. [12] found that in MUAs with at least one FQHC, patient volume was 3.27 times higher than in MUAs with none, and that FQHC placement patterns may reflect historic redlining. These patterns suggest a long-standing mismatch between where facilities are placed and where they are most needed.

2 Our Approach

The problems described above share a common structure: they involve making spatial decisions under constraints such as budget, capacity, travel time, equity, with many interacting pieces. This is what optimization models do: evaluate many configurations and find good ones. We propose a spatial optimization framework that addresses the HRSA requirements listed above.

2.1 A hierarchical network design via FalCom

FalCom [10] is a Markov chain Monte Carlo method from applied math designed for hierarchical and capacitated facility location and districting problems. The energy of a district measures how well a facility within that district serves its demand points. The system seeks to equalize energies across districts: the sampler favors configurations where no district is left with disproportionately high energy. The energy function is user-defined and can serve as an access measure, incorporating any combination of factors such as population-provider ratio, travel time, insurance coverage, or chronic disease burden. Given a set of facilities, demand units, and an energy function, FalCom generates a collection of high-quality spatial configurations, each assigning demand units to facilities and grouping facilities into higher-level clusters, rather than returning a single optimal solution. This allows decision makers to compare alternative configurations and choose based on their priorities. The model organizes Chicago’s healthcare landscape into two levels, matching the structure HRSA requires but has no tools for:

Districts are catchment areas for primary care, the spatial units that HRSA’s **R1** (catchment area design) requires each health center to define. Each district groups a set of census tracts around one FQHC or clinic. Today, health centers draw these boundaries case-by-case, without a formal methodology; our model draws them optimally. We construct boundaries so that:

- **Catchment area design.** District boundaries are geographically contiguous and respect residential patterns, economic groupings, and available transportation, satisfying the Catchment Area Design requirement **R1** in the Compliance Manual. The model determines the optimal grouping of census tracts into catchment areas, exactly the spatial decision HRSA currently leaves to applicants with no analytical support.
- **Needs assessment.** Each district maintains demand-capacity balance, i.e., the ratio of population need to available provider capacity is properly adjusted, satisfying the Needs Assessment requirement **R2** in the Compliance Manual. The district’s energy value corresponds to its IMU score, linking the optimization directly to HRSA’s underservice metric.
- **Accessible locations.** Every resident is assigned to a facility based on an access function $\text{Access}(x, y)$, where x is a facility and y is a demand unit. Travel time by CTA transit is

one component; the framework supports any definition of access (e.g., travel time, distance, provider availability, or a composite). For any such definition, the model constructs compact and equitable districts, satisfying the Accessible Locations requirement **R3** in the Compliance Manual.

Superdistricts group multiple primary care districts around a referral hospital. Each superdistrict ensures that every clinic within it has a clear referral path to hospital and specialist care. Superdistricts are designed so that:

- **Hospital referral relationships.** Each primary care site is linked to a hospital its patients can reach, satisfying the Hospital Referral Relationships requirement **R4** in the Compliance Manual. The superdistrict energy measures referral connectivity, i.e., how well patient flows stay within the superdistrict for specialist care.
- **Collaborative relationships.** Clinics within a superdistrict share a referral hub, strengthening the Collaborative Relationships requirement **R5** in the Compliance Manual and reducing fragmented care and non-urgent emergency department use. Fishman et al. [8] found that in Chicago, patients in areas with lower spatial access to primary care had significantly higher odds of preventable emergency department use.

FalCom is a sampling method, not a single-solution optimizer. It generates many good configurations for decision makers to compare and choose. It creates districts and locates facilities in a way that balances energy values across the system as much as possible.

2.2 Additional model components

Transit-based travel time. We use CTA public transit travel times, not straight-line distance, as the access metric, directly addressing Ch. 6’s requirement to consider distance and time taken for patients to travel. In Chicago, 36.9% of African American households lack a vehicle [7]. Ermagun and Tilahun showed that 50% of African Americans in Chicago live in the lowest third of transit accessibility areas, compared to 21% of whites. Milano [14] found that only 8% of at-risk populations in New Jersey could reach an FQHC within 15 minutes by bus, regardless of time of day. A clinic 2 miles away may require a 45-minute bus trip. Our model uses the actual transit network so that *close* means close for the people who need the care.

Demographic balance. Balance across superdistricts can be enforced, for example ensuring no superdistrict has a disproportionately high uninsured rate or chronic disease burden.

Redesigned underservice scoring. We will develop data-driven alternatives to the binary IMU ≤ 62 cutoff. Instead of a hard threshold that labels a census tract as either underserved or not, our approaches produce continuous measures that capture degrees of need. We are exploring three directions:

1. **Probabilistic IMU.** Two of the four IMU indicators carry substantial uncertainty. Poverty rate and proportion aged 65+ come from the American Community Survey, where in approximately 50% of census tracts the margin of error exceeds 50% of the estimate itself [21]. Instead of computing a single deterministic score, we model each indicator with its uncertainty distribution and produce a *probability of being underserved* for each tract. A tract with IMU point estimate 63 but high variance might have a 40% probability of actually falling below

62. This replaces the binary cutoff with a continuous risk measure and naturally handles measurement error.
2. **Census block-level IMU projection.** The current system computes IMU at the census tract or rational service area level, which masks within-area variation. We project IMU components down to the census block level using 2020 Decennial Census block data (population, age 65+) and ACS tract-level estimates (poverty, provider counts). This reveals where the sharp cutoff actually falls spatially: a tract scoring 63 (not designated) may contain blocks that are severely underserved but invisible to the current system.
 3. **Expanded scoring for optimization.** Our model can incorporate factors the 1970s IMU formula omits: insurance coverage rates, CTA transit travel time to the nearest provider, chronic disease burden (diabetes, hypertension prevalence), and historical redlining indicators. Rather than proposing a single replacement formula, we let the optimizer use these as inputs: the decision maker specifies which factors matter and how much, and the model finds facility configurations that optimize across them. This separates the policy question (what should count as underserved?) from the technical question (given a definition, where should facilities go?).

3 Current Status

3.1 Data

We have assembled a detailed spatial inventory of Chicago’s primary care infrastructure, integrating six federal and commercial data sources:

- **166 FQHC sites** (HRSA UDS), cleaned from 213 by removing non-primary-care outreach, shelters, and admin offices
- **128 hospitals** with certified bed counts (CMS Provider of Services, Q4 2025)
- **416 private clinics and urgent care** (Google Places), deduplicated against federal sources
- **61,460 individual providers** mapped to facilities by practice address (CMS NPPES, Feb 2026)
- **35 MUA designations** with IMU scores (HRSA MUA Find); **33 health indicators** across 77 community areas (Chicago Health Atlas [5])
- **1,363 HRSA grant awards** for Cook County (2016–2026)

Provider matching revealed that 34 of 166 FQHC sites have zero providers registered at their address, indicating extremely thin staffing or data gaps that themselves signal capacity problems.

3.2 Dashboard

We have been developing an interactive decision-support dashboard coded in JavaScript. A demo version and all data sources are publicly available.¹

¹<https://akirtisoglu.me/research/chicago-healthcare-network>

The dashboard allows users to explore Chicago’s healthcare landscape at multiple geographic scales: census blocks, tracts, community areas, and health zones. Users can toggle facility layers (FQHCs, hospitals, private clinics), view provider counts and specialty breakdowns per site, overlay MUA boundaries with IMU scores, and examine 33 health indicators including life expectancy, insurance rates, and chronic disease prevalence. Health indicator data is sourced from the Chicago Health Atlas [5].

Planned additions include redesigned IMU-like underservice scores (Section 2) and an integrated FalCom sampling module that will allow users to generate and compare district-superdistrict configurations directly within the dashboard. Screenshots of the current dashboard are included in the Appendix.

3.3 What We Need

Data.

- *Per-site UDS data* (visits, FTEs): HRSA publishes UDS data only at the organization level, not per site. A data use agreement with HRSA would unlock site-level capacity data.
- *CMS referral data post-2015*: The Physician Shared Patient Patterns dataset covers 2009 to 2015. Newer data may require a ResDAC application² or FOIA³ request; university affiliation strengthens these.

Connections.

- Prof. Kim at UIC, whose work [11] is the main source of motivation for Gap 3, could provide guidance on MUA designation gaps and South Side access disparities.
- A connection with UChicago would be valuable to discuss FQHCs on the ground and design the project more realistically toward community partnerships in the future.
- Introductions to FQHC network operators, Chicago Department of Public Health, or Illinois Primary Health Care Association to validate data.

Potential funding opportunities.

- *NSF Smart & Connected Communities*: Up to \$2.5M/4yr. Requires community partner. Planning grants (\$150K) could fund pilot work with CDPH. Preliminary proposals required; see NSF 25-527 for 2026 dates.
- *NSF Smart Health (SCH)*: Up to \$1.2M/4yr (joint with NIH). Computational methods for healthcare. Next solicitation cycle expected late 2026.
- *AHRQ Primary Care Research (R01)*: Up to \$500K/yr, 5yr. Explicitly supports primary care access and equity research. Rolling deadlines: February 5, June 5, October 5 each year.
- *NSF CMMI Operations Engineering (OE)*: \$200K–\$500K over 3 years. Core disciplinary home for location-allocation, resource allocation, and public-sector decision models. Continuous submission, no community partner required.

²ResDAC (Research Data Assistance Center): <https://resdac.org/>

³Freedom of Information Act: a legal right to request unreleased federal agency records.

- *RWJF Health Equity Scholars for Action*: Up to \$260K per award. For early-career investigators within 10 years of terminal degree. RWJF is launching a new health equity research initiative in 2026, replacing its current signature programs.

3.4 Roadmap

Phase 1: Publication (Current – May 2026)

- Reframe the narrative with Prof. Erwin
- Discuss a full model with Prof. Kaul
- Implement FalCom and submit the paper

Phase 2: Tool Development and Follow-up Paper (2026–2027)

- Present the paper at UChicago, UIC, INFORMS Health Applications
- Submit funding application for a full implementation
- Integrate the analysis module (select location → see impact) into the dashboard
- Work on a follow-up paper: Chicago Physician Referral Network (Section 4)

4 Future Work: Chicago Physician Referral Network

Chicago has one of the largest FQHC networks in the country including major academic medical centers such as UChicago, Northwestern, Rush, and UIC, yet no one has mapped how referrals actually flow between primary care and specialists in the city.

An et al. [2, 1] constructed the national physician referral network from the CMS Physician Shared Patient Patterns dataset, where physicians are nodes and shared Medicare patients within 30-day windows form edges. The national network contains over one million physicians and 55 million referral records. They found strong hub-and-spoke structure with 93% of referral steps staying within the same hospital region, and referral volumes following a gravity law inversely proportional to distance. CMS made this data publicly available between 2009 and 2015 but has not released it since. A follow-up paper could request current data from CMS for 2025 and construct a Chicago physician referral network, overlaying it with our FQHC and hospital data to identify referral deserts where patients have no viable path from primary care to specialist care. Obtaining the data would require a Research Data Assistance Center (ResDAC) application or a data use agreement with CMS. A collaboration with UChicago or UIC, both of which have existing CMS data agreements, could make accessing this data significantly easier.

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Appendix: Dashboard Screenshots

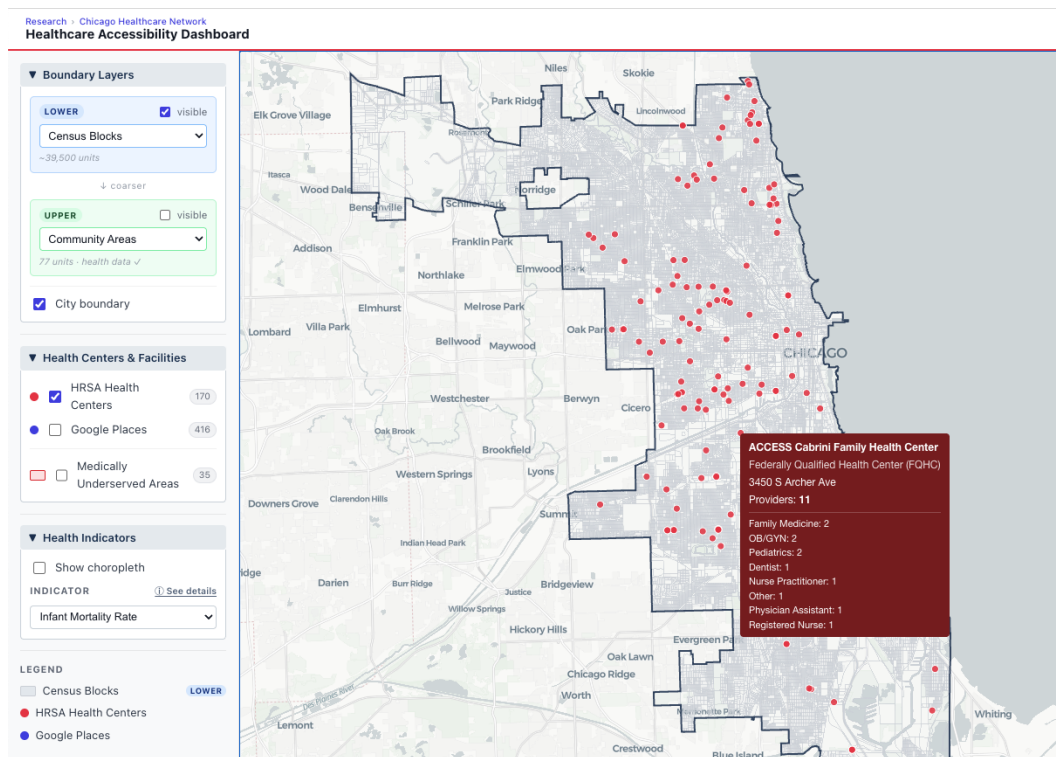


Figure 1: Census blocks within the city boundary. Red markers are FQHC sites. The tooltip shows the number of practitioners in the selected center.

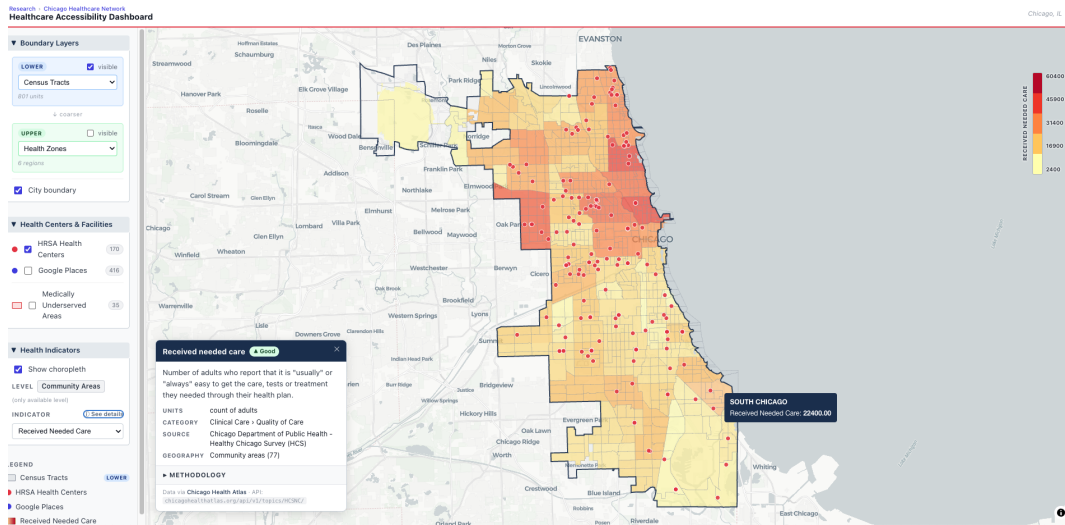


Figure 2: Health indicator overlay (Received Needed Care rate) at the census tract level, with FQHC sites and a community area tooltip for South Chicago showing local statistics.

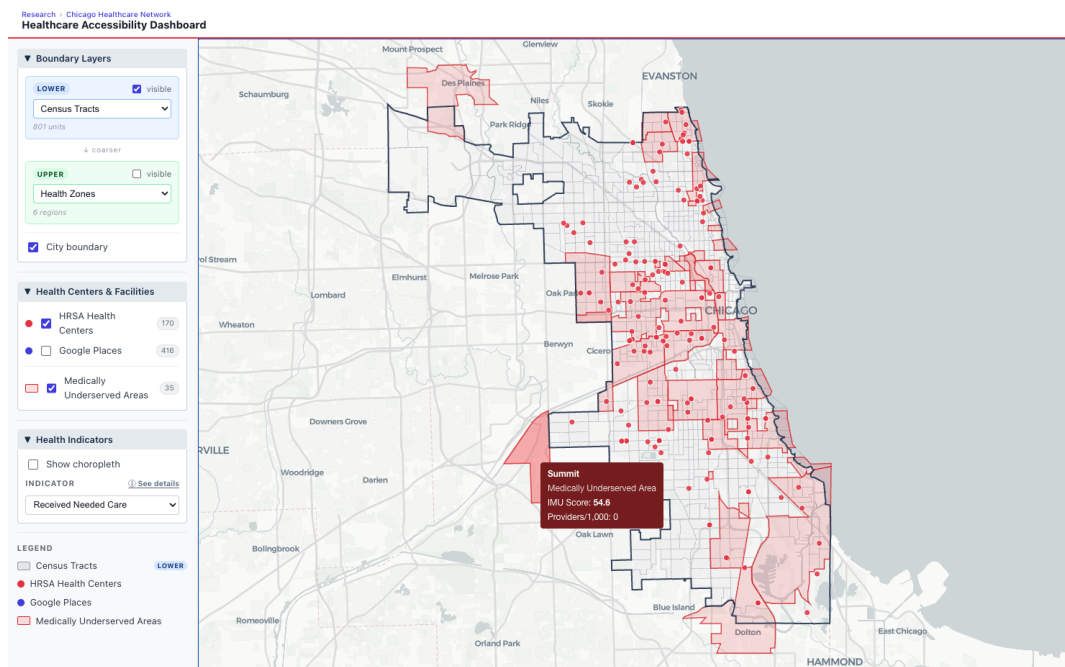


Figure 3: MUA boundaries with FQHC sites. The tooltip shows the IMU score of Summit as 54.6.

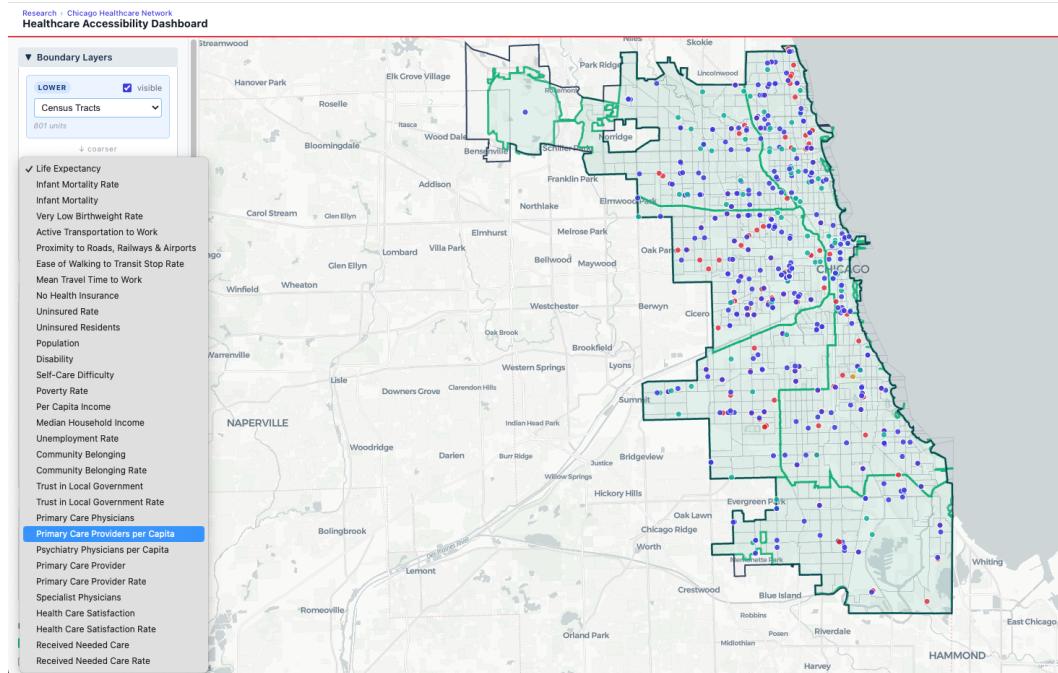


Figure 4: Health zones (upper level) and census tracts (lower level) with all facility layers: FQHC sites (HRSA), hospitals, and Google Places primary care and urgent care locations. The two datasets share common facilities and require deduplication. The panel shows the full menu of 33 health indicators available for overlay, including life expectancy, infant mortality, insurance rates, chronic disease prevalence, and primary care provider rates.